Conditional Generation from Few-shot Examples for Protein Design

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6.s981: AI for Protein Biology

The Challenge of Protein Design

- **Goal:** design proteins with specific functional properties (e.g., therapeutic proteins, industrial catalysts)
- **Current Obstacles:**
  - **Vast Search Space:** known proteins are a tiny fraction of possible sequences
  - **Rugged Landscape:** relationship between sequence + function is complex
  - **Data Scarcity:** experimental synthesis is limited; labeled data for specific functions (ex. enzyme classes) is sparse
- **Problem with Existing Models:** standard PLMs require fine-tuning on large labeled datasets, which is difficult for understudied or novel enzyme families

Landscape of Protein Design and Learning

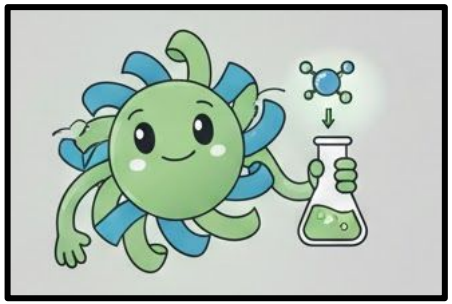
- **De Novo + Generative Design**
  - Traditional physics-based methods (e.g., Rosetta) are accurate but computationally expensive
  - DL models (GANs, ProtDiff) often require task-specific FT or explicit labels
- **Protein Language Models**
  - Models like ESM2 & MSA Transformer capture evolutionary constraints
  - Lack effective conditioning mechanisms for specific functional design without retraining

Bridging the Gap

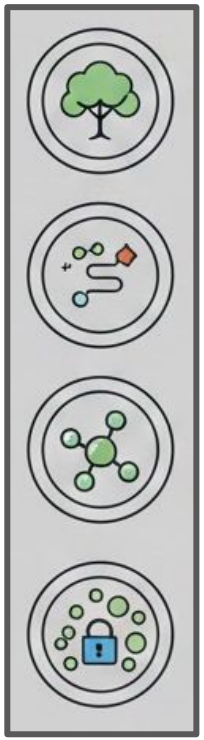
- **ML for Enzymes**
  - Current methods (e.g., EC2Vec, GraphEC) focus on classification and prediction rather than generation
  - Data scarcity is a bottleneck
- **In-Context Learning**
  - Borrowing from LLMs (e.g., GPT-3), ICL adapts to new tasks using just input examples
  - This project introduces ICL-style module to PLMs to aggregate information across few-host protein examples

Enzyme Commission (EC) System

EC Number System classifies enzymes by reaction types



Most enzymes are proteins that catalyze chemical reactions



First level *x*: 7 Classes
1: Oxidoreductases

Second level *x.y*: 79 Subclasses
1.1: Acting on the CH-OH group of donors.

Third level *x.y.z*: 322 Sub Subclasses
1.1.1: With NAD(+) or NADP(+) as acceptor.

Fourth level *x.y.z.w*: 8,309 Serial numbers
1.1.1.1: Alcohol dehydrogenase

Broad chemistry

Specific Catalysis

Proposed Solution: Few-Shot Generative Framework

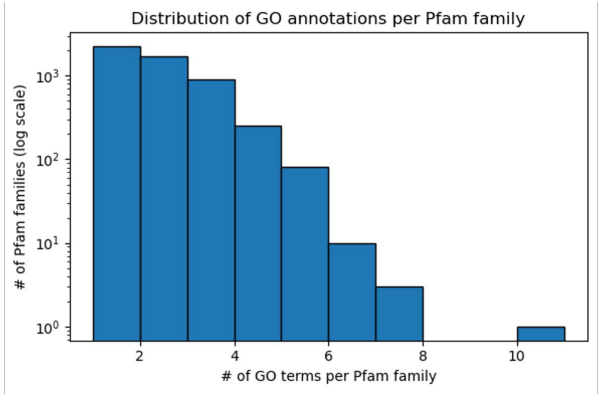
- **Core Concept:** generative model that learns from a few examples rather than large datasets
- **Methodology:**
  - **In-Context Learning:** conditions on a small set of representative sequences that implicitly define a functional context
  - Model Architecture: Extend pre-trained MSA Transformer with a diffusion-based generative objective
- **Key Innovation:** use Classifier-free Guidance (CFG) to steer generation towards desired features without relying on explicit functional labels (like EC numbers) or external scorers

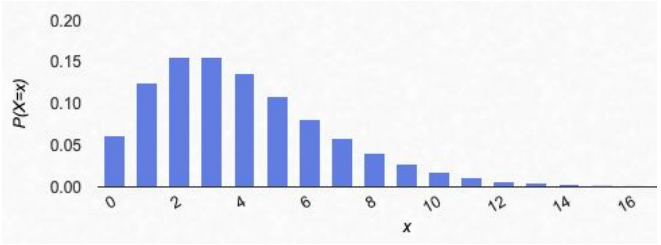
2-stage pipeline:

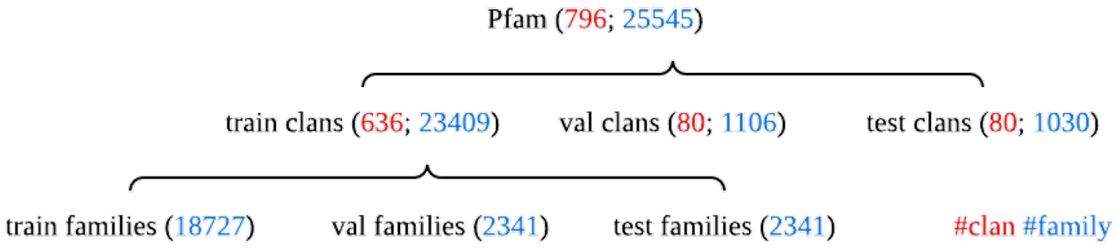
1. Learn functional constraints from a few sequences (training)
2. Extrapolate at inference using CFG

Dataset:

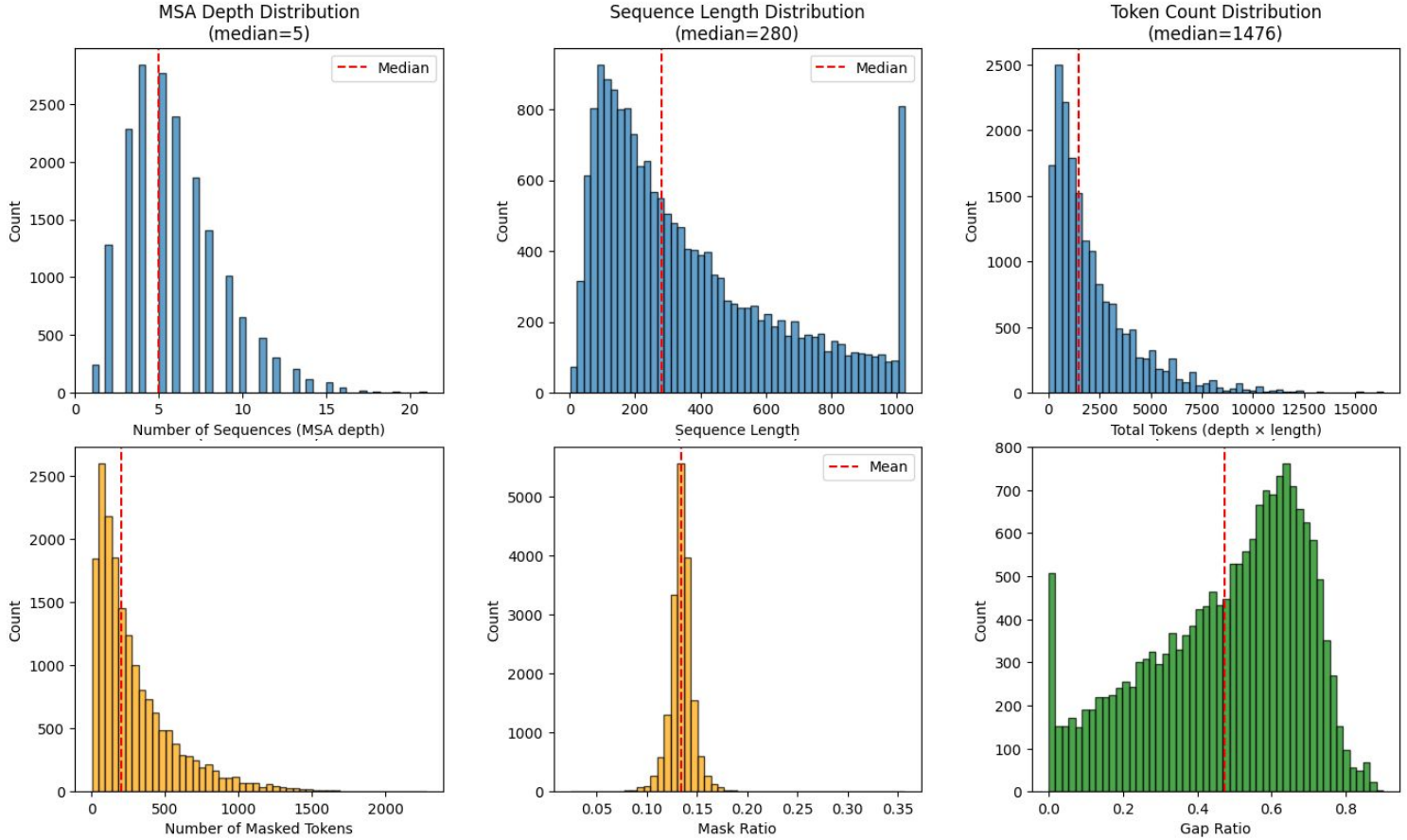
- Source: Pfam
  - Families: multiple sequence alignments (MSAs) constructed using hidden Markov models
  - Clans: families grouped by sequence and structural similarity
  - Within-family Functional Homology (# of distinct GO labels)
- Pre-processing
  - Dataset split
  - Subsampling: $(K-1) \sim NB(r=4, p=0.5)$
    - Average depth around 5
    - Maximum depth = 20





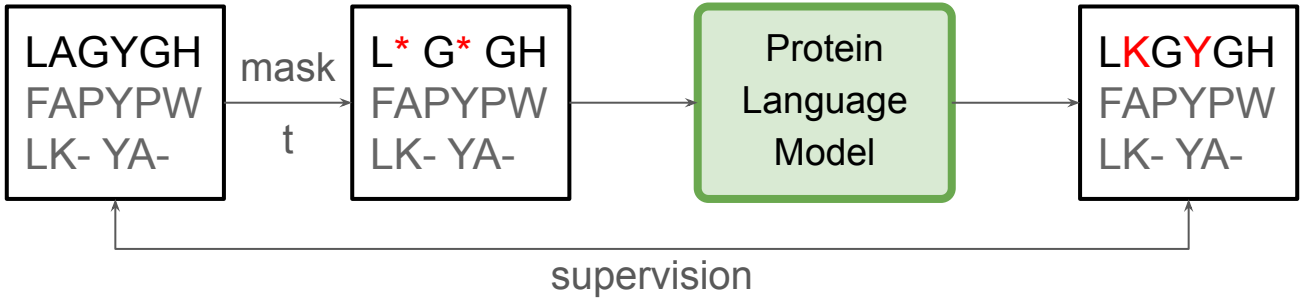


Pre-processed Dataset Overview:



Training: Learning Constraints from Few Examples

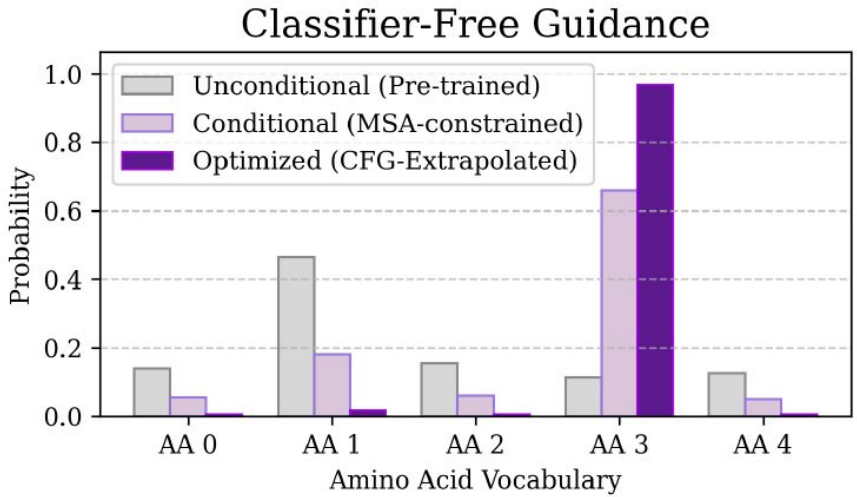
Formally, an MSA is represented as a matrix $x \in \mathbb{R}^{D \times L}$, where each row $x^{(i)}$, $i \in \{0, \ldots, D-1\}$ corresponds to an individual sequence and columns correspond to residue positions. Conditioned on the context rows $x^{(1:D-1)}$, the model aims to reconstruct the target sequence $x^{(0)}$.

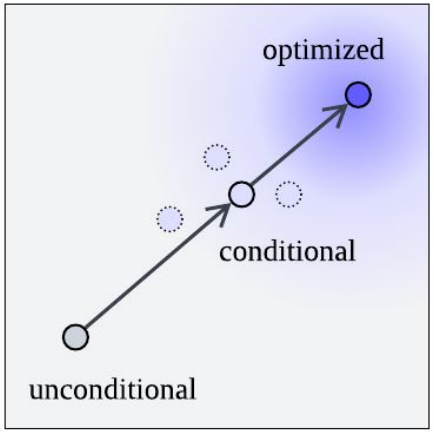


$$\mathcal{L}(x; t, \theta) = \sum_{i \in \mathcal{M}_t} \log p(x_i^{(0)} \mid \tilde{x}^{(0)}, x^{(1:D-1)}, t, \theta)$$

Inference: feature optimization via classifier-free guidance

$$\text{logits} = \text{logits}_{\text{uncond}} + \lambda(\text{logits}_{\text{cond}} - \text{logits}_{\text{uncond}})$$





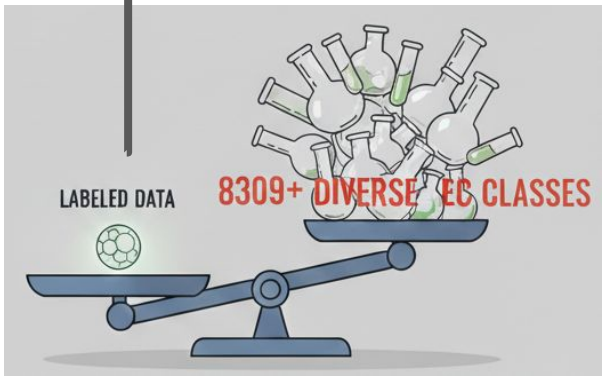
$$\text{logits} = \text{logits}_{\text{neg}} + \lambda(\text{logits}_{\text{pos}} - \text{logits}_{\text{neg}})$$

Evaluation: Benchmarks and Metrics

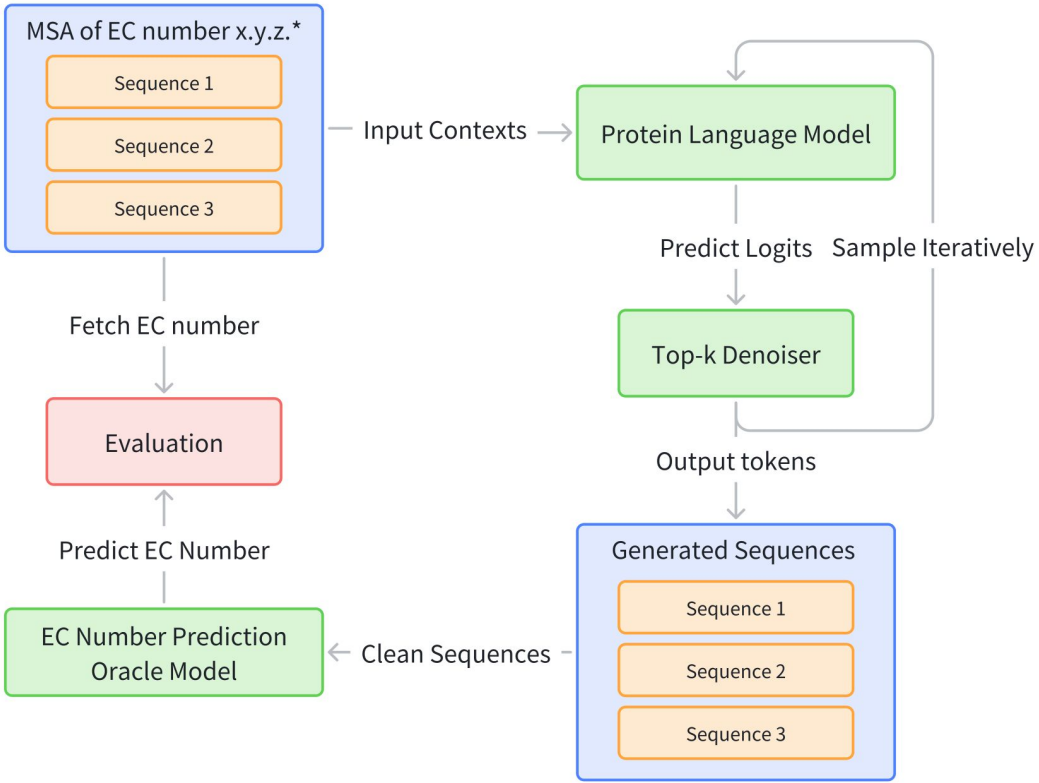
1. **Conditional Enzyme Generation**
    - Dataset: Price-149, Rare Enzyme in UniProt
    - Metrics: success rate evaluated by EC number prediction oracles
2. **Fitness Score Prediction with Classifier-Free Guidance**
    - Dataset: ProteinGym (Deep Mutational Scanning)
    - Metrics: Spearman Correlation, Top-5 Success Rate

Conditional Enzyme Generation

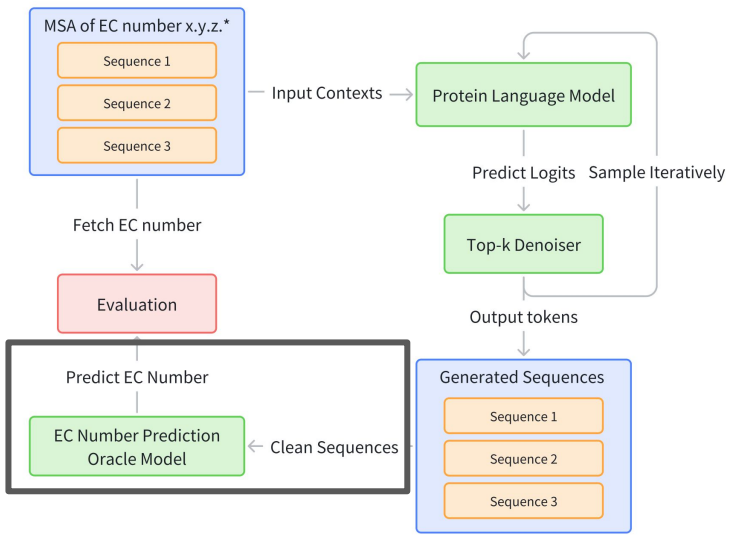
Few shot in-context learning



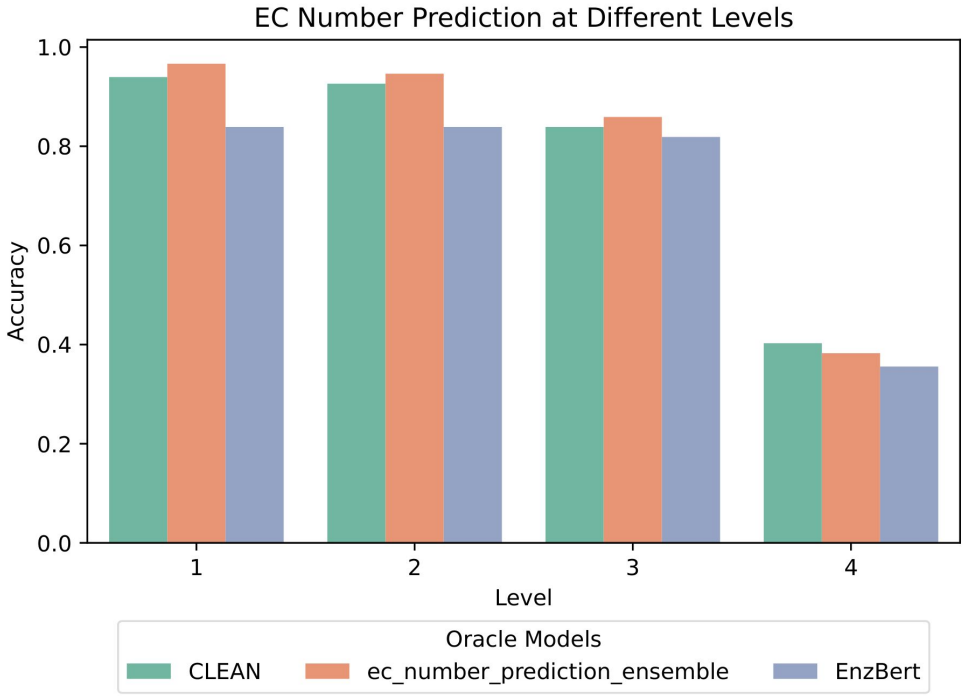
Price-149, UniProt selections



Conditional Enzyme Generation: Oracles

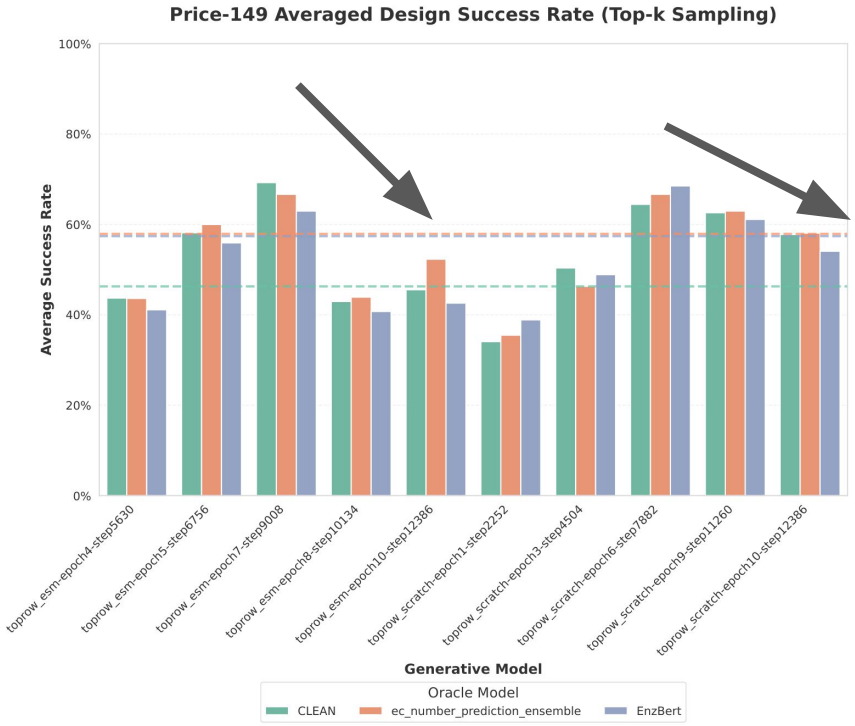


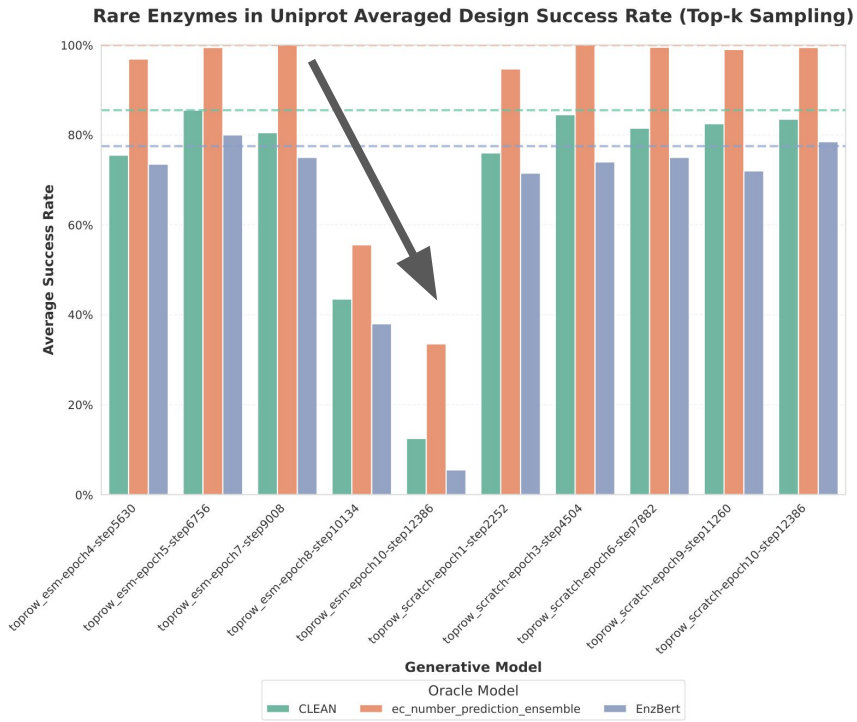
Computational surrogate of design evaluation



SOTA models: decent reliability on *x.y.z* level

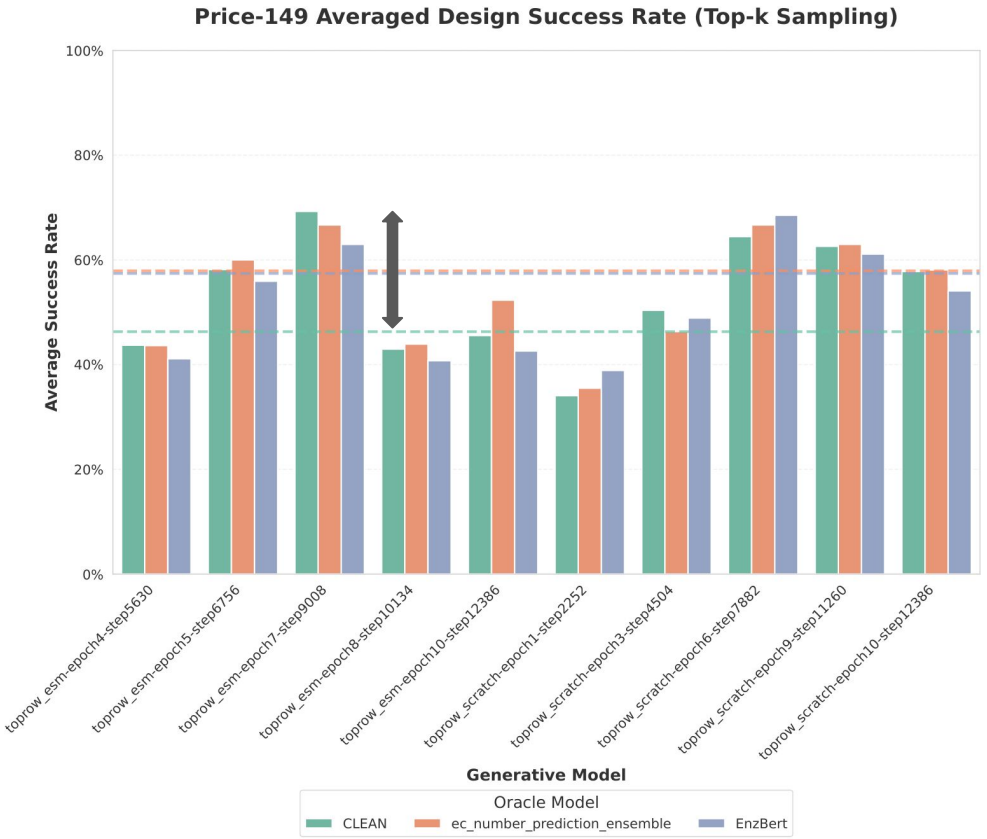
Conditional Enzyme Generation: Overfitting





Best models appear before 10 epochs full training

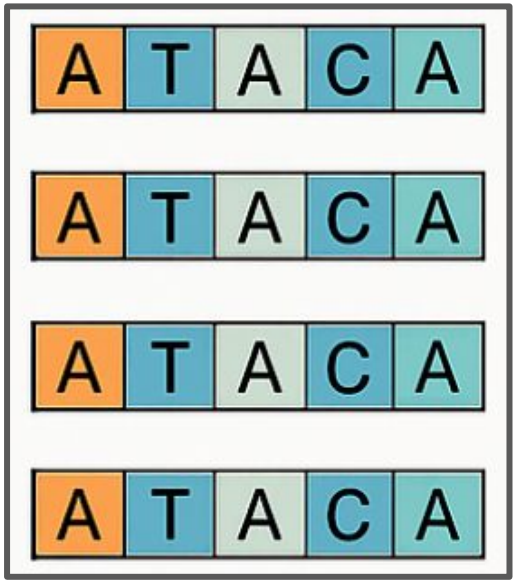
Conditional Enzyme Generation: In-context Learning



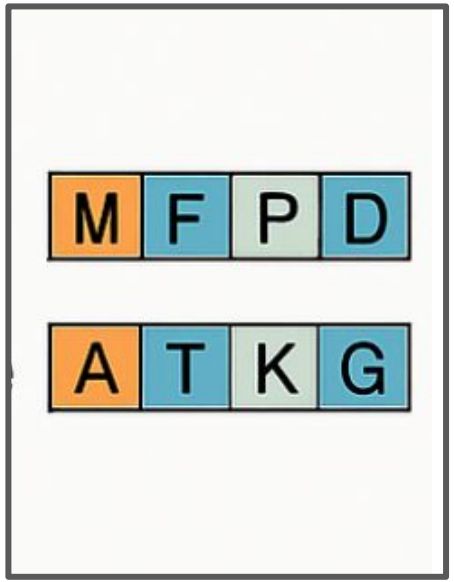
Performance gain against random mutation baseline with the same mutation rate as the best model's design diversity to the contexts

Conditional Enzyme Generation: Reflection

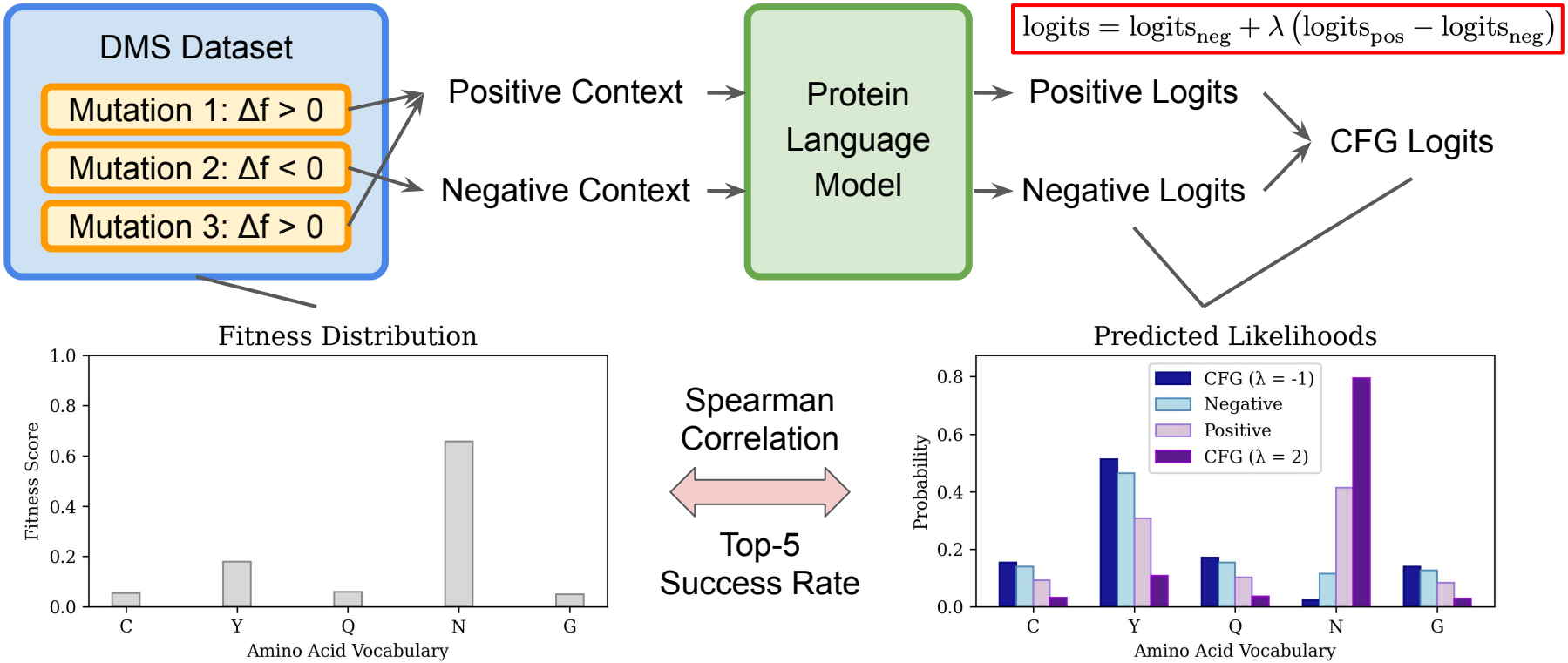
Typical MSA scenario: high homological relevance and sufficient depth



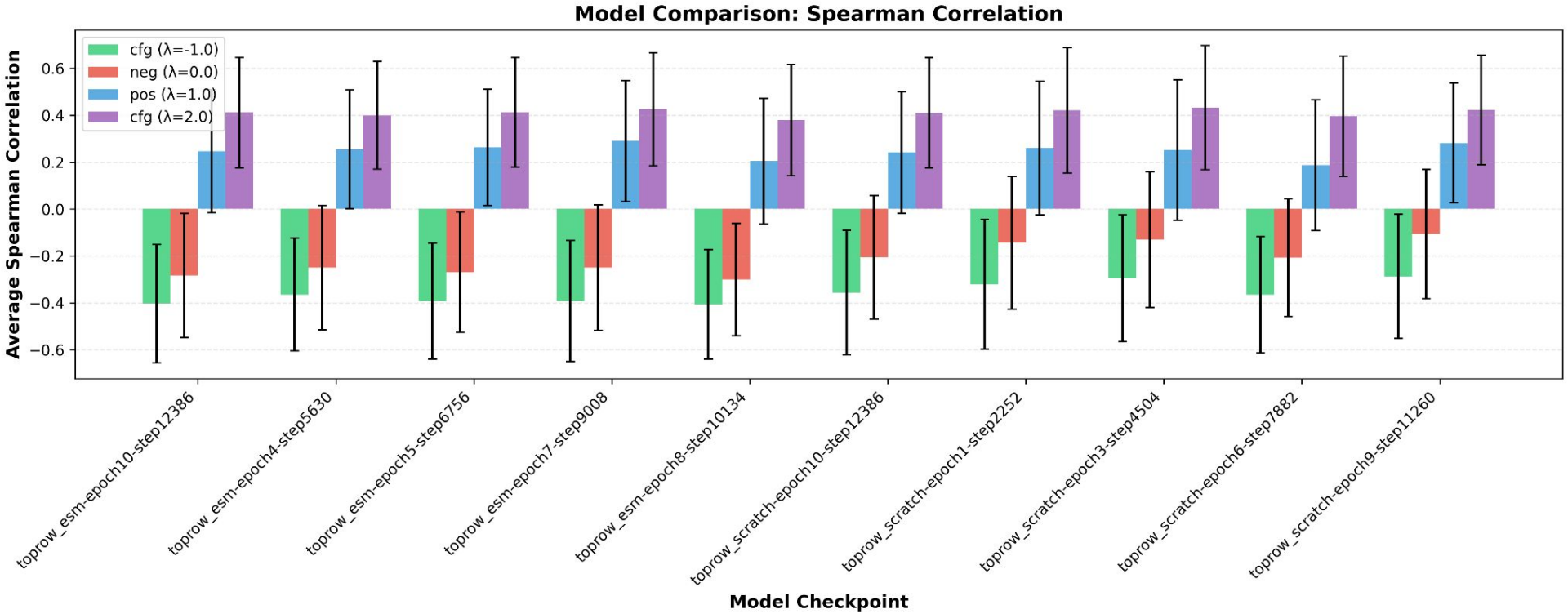
Enzyme design scenario: shallow and diverse sequence pools with the same function



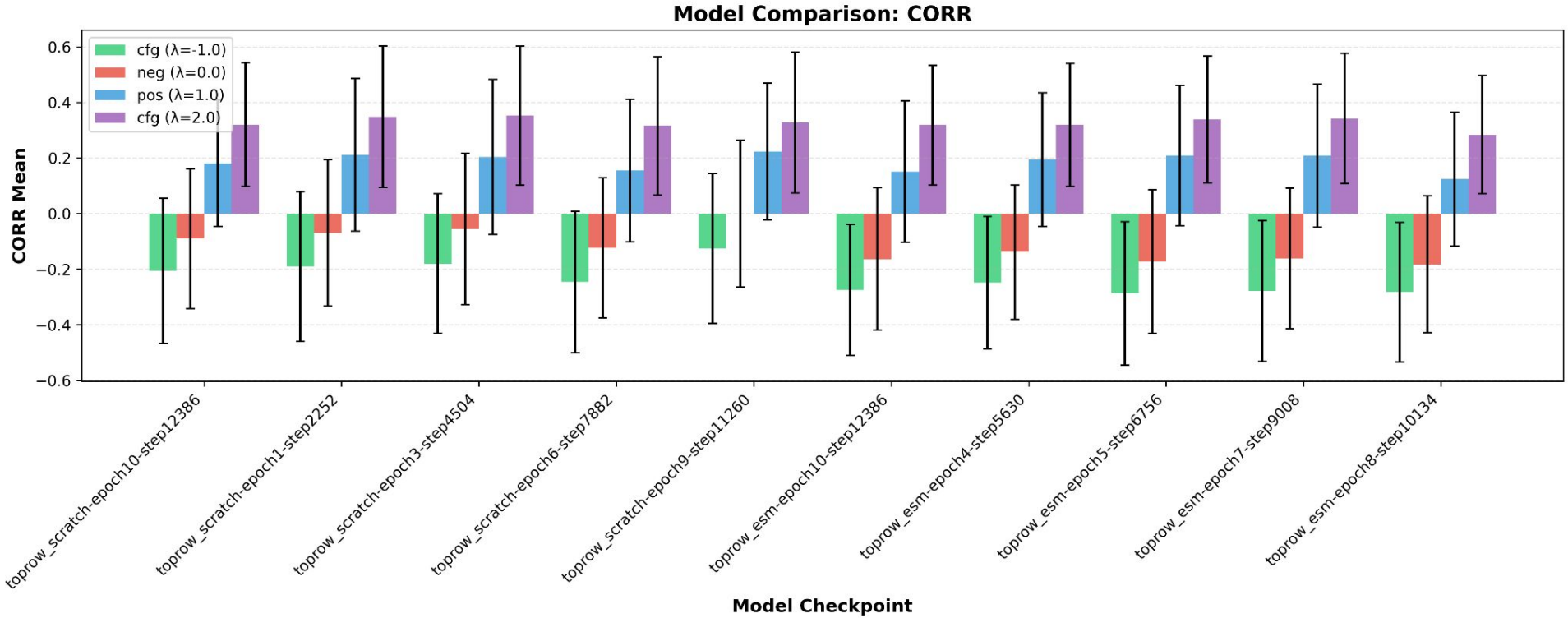
Fitness Prediction with Classifier-Free Guidance



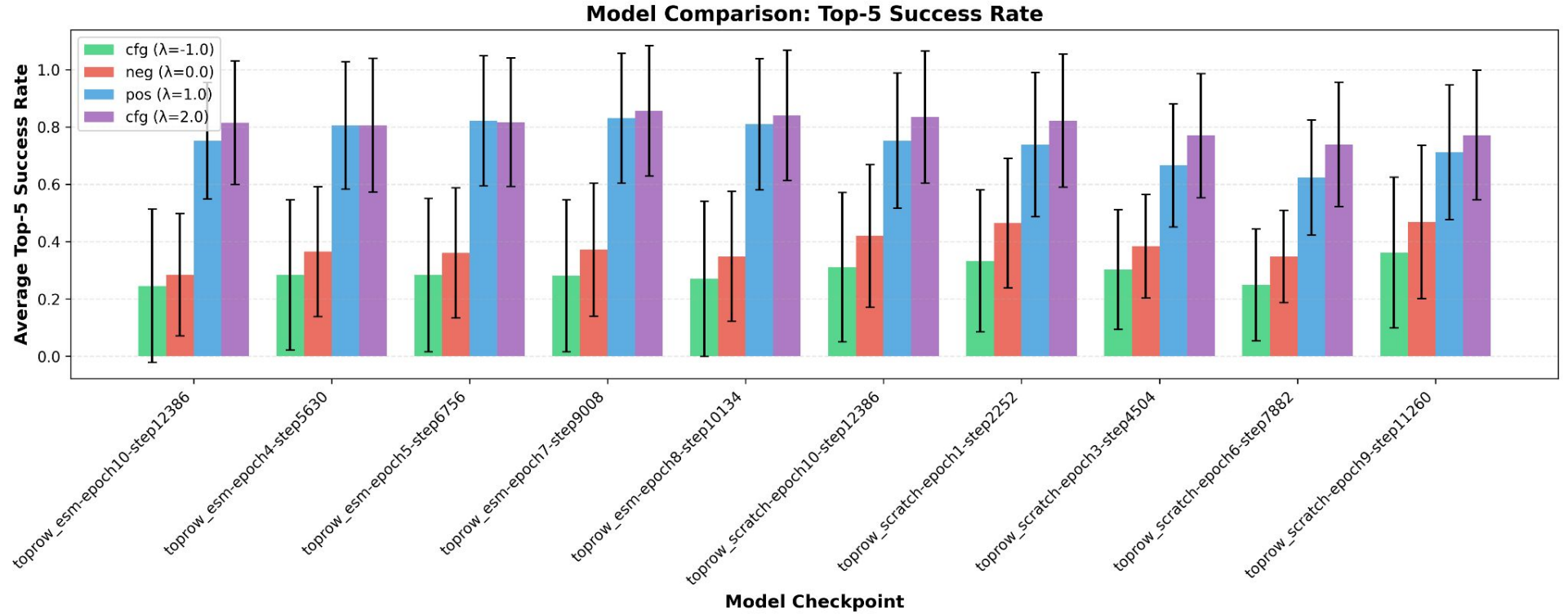
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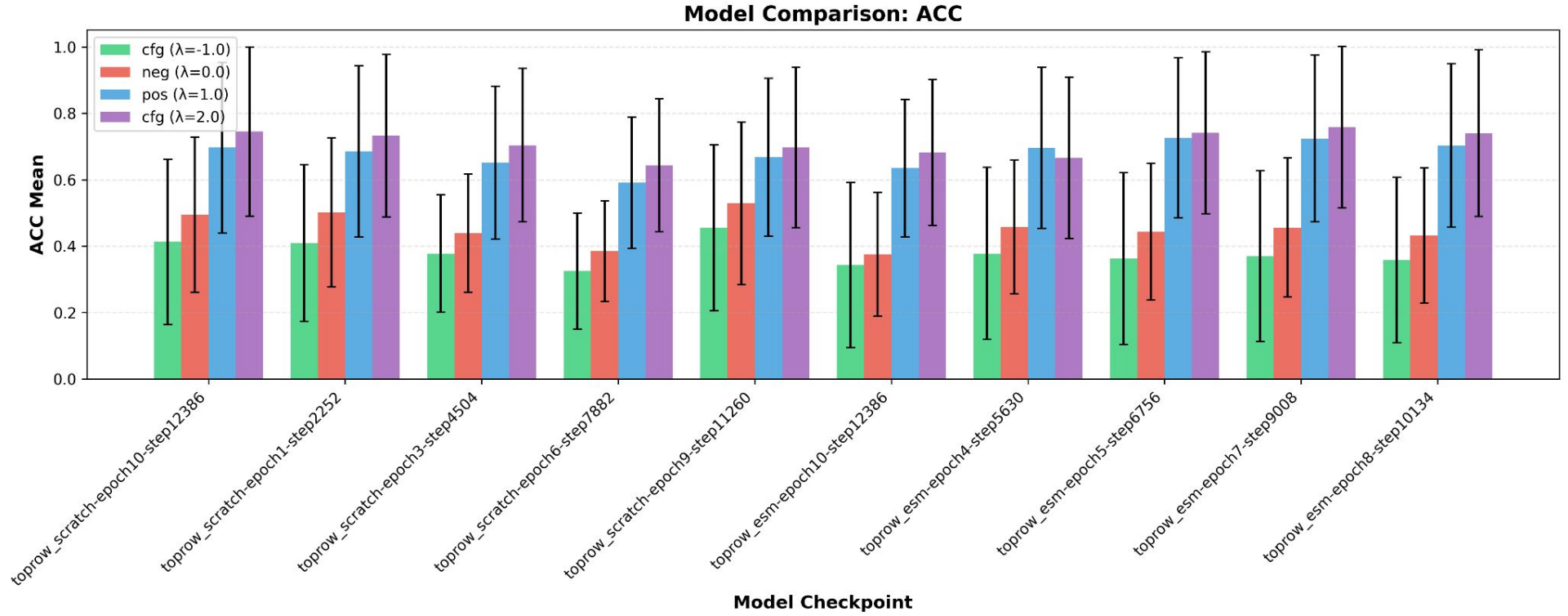
Fitness Prediction with Classifier-Free Guidance



Fitness Prediction with Classifier-Free Guidance



Fitness Prediction with Classifier-Free Guidance



Fitness Prediction: Takeaways

- Conditioning Effect
  - The model successfully infers fitness-relevant signals from context
- Impact of Guidance
  - Increasing guidance magnitude (λ) proportionally improves Spearman correlation
  - Top-5 success rates show modest improvement
  - Guidance reshapes the less-confident section of the likelihood distribution
- Robustness across Model Checkpoints
  - Extracting fitness signals from context is a capability the model acquires easily

Key Takeaways

- **Success in Low-Data Regimes**
  - Model can generate functional sequences using only 10-30 examples
  - Achieves 50-90% design success rate across 3 independent oracle predictors
- **Effective Conditioning**
  - ICL works: conditioning on functionally related sequences improves fitness prediction over random baselines
  - Pretrained MSA Transformer encodes sufficient evolutionary priors to extrapolate from shallow contexts
- **Inference Strategy:**
  - Top-k sampling (k=16) outperforms confidence-based sampling
- **Classifier-Free Guidance**
  - Separates high & low fitness substitutions without modifying model parameters

Key Takeaways

- Propose a novel diffusion-based few-shot functional conditioning method
- No labels needed → just example sequences
- Strong functional generalization across enzyme families
- Zero-shot fitness prediction boosted by classifier-free guidance

Future Work

- Evaluate novelty (sequence and structural diversity) of generated sequences
- Incorporate structural constraints (3D models)
- Experimental validation

Thank You!